TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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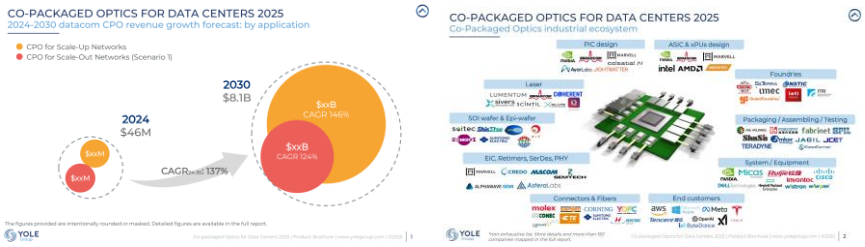
2. Technology Overview

Title of Proposed Technology	3D optical transceiver using optical-waveguide embedded glass interposer for AI server communications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	ICT Modules and Components	Optical Communication Modules and Components	Optical Application Devices	Optical Components
Concept of Proposed Technology	The proposed technology proposes a 3D optical transceiver using optical-waveguide embedded glass interposer for AI server communications. ● **Key Innovations** 1. 3D Integrated optical transceiver: ✓ Ultra-compact formfactor and low-power optical transceiver ✓ 3D packaging of optical and electrical components ✓ Next generation co-packaged optics 2. Optical-waveguide embedded glass interposer: ✓ Multi-layer RDL glass interposer ✓ Optical waveguide embedded ✓ Support for fiber-coupling structure (gradient-index surface coupling)			

Rationale for Joint R&D Project

- **Technical Necessity**
  - 최근 광 interconnect 기술은 종전 telecomm 또는 data comm 분야를 넘어 AI data center의 서버간 통신에 사용될 것으로 확실시 됨.
  - 광전 소자 트랜시버 기술은 legacy 기술임에도 불구하고, 패키징 및 테스트 가격, supply chain의 능동성등의 이슈로 높은 가격에 형성되어 있음.
  - 최근 CPO (co-packaged optics) 등의 기술을 중심으로 신호의 완결성 및 고속성을 이루기 위해 업계를 중심으로 많은 활동이 진행 중임.
  - CPO에 사용될 차세대 Glass interposer 기술은 최근 큰 각광을 받고 있으며, 광도파로가 집적된 glass interposer 기술을 이용한 광송수신 모듈 개발은 반드시 필요한 기술임
- **Commercial and Strategic Drivers**
  - 시장 조사 기관인 Yole Group에 의하면, 전세계 CPO 시장은 2024년 $46M 시장이었던 것으로 추정이 되며, CAGR 137 %의 대단히 빠른 속도로 발전을 할 것으로 예상됨. 2030년 전세계 CPO 시장은 약 $8.1 B으로 예상되고 있음
  - 이러한 산업적으로 큰 각광을 받으며, CPO 기술을 위한 전세계 생태계는 대단히 빠른 속도로 전개되고 있음



Co-packaged optics market (left) and industrial ecosystem (right) (source: YOLE Group, Co-packaged Optics for Data Centers 2025)

- **Development Timeline and Cost Optimization**
  - Joint R&D with Fraunhofer Institute reduces development efforts (infrastructures for fabrication and characterization, etc.) considerably, approximately 40-50% by a whole.
  - According to survey, korean partner(s) alone cannot achieve optical-waveguide embedded glass interposer technologies within the project timeline.
- **International Collaboration Benefits**
  - Secured access to Fraunhofer's proprietary glass substrate fabrication processes and characterization infrastructure
  - Enhanced global competitiveness and technology visibility through association with a world-class research institute
  - Accelerated commercialization pathway through European market channels and global certification frameworks

Key Words

1. Optical waveguide embedded multi-layer glass Interposer
2. 3D packaging
3. optical transceivers

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	● **Primary Objective** Develop and demonstrate 3D optical transceiver using optical-waveguide embedded multi-layer RDL glass interposer for AI server communications. (RDL: Re-Distribution Layer) ● **Specific Technical Objectives** 1. Glass Interposer integrated with optical waveguides ✓ Design and fabricate multi-layer glass substrates with optimized optical waveguides ✓ Implement through-glass vias (TGV) and multi-RDL enabling flip-chip IC integration 2. 3D packaging of opto-electronics components ✓ Optical modulators and photodetector developments ✓ Implement a 3D packaging ✓ Test at > 32 Gbps			
R&D Target Specifications	1. Integrated optical transceiver onto optical waveguide integrated glass interposer 2. Photodiode, Transimpedance amplifier 3. Mach-Zehnder modulator, driving ASIC 4. Operation speed: > 32 GBd			
R&D Plan	● **Phase 1: Feasibility Verification (6 Months)** ■ Objective: Demonstrate proof-of-concept for 3D integrated optical transceiver using optical waveguide embedded glass interposer ■ Key Tasks: ✓ Define detailed system requirements and target specifications ✓ Simulation of optical waveguide in glass interposer ✓ Define glass interposer design rule (number of RDL, packaging methodologies, etc.) ● **Phase 2: Prototype Development and packaging (30 Months)** ■ Objective: Develop and validate a TRL 5-level prototype of a optical transceiver using optical waveguide embedded glass interposer ■ Key Tasks: ✓ Fabricate and characterize optical waveguide embedded glass interposer ✓ Flip-chipping of optical components and ASICs (TIA, driver) ✓ Demonstrate optical transceiver performance, > 32 Gbps ✓			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	■ Fabrication of optical waveguide embedded multi-layer glass interposer ✓ Direct glass substrate design, fabrication process development, and characterization ✓ Lead TGV drilling, RDL patterning, and multilayer integration technology ■ Driver, TIA design and fabrication ✓ Driver ASIC for Mach-Zehnder modulator
Expected R&R of Korean SME Partner	■ 3D packaging of glass interposer with ASICs ✓ Flip-chip packaging of components ✓ Optical fiber alignment ■ Prototype module test ✓ Test platform ✓ Module test at > 32 GBd (TBD)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● **Technological Achievements** ■ Direct Outcomes ✓ World-best ultra-small formfactor optical transceiver exploiting optical waveguide integrated glass interposer ✓ Low-cost mass production of optical transceiver for AI data centers ✓ Prototype development ■ IP and Knowledge Outcomes ✓ High-impact patent applications ✓ Peer-reviewed publications ✓ Design documentation, technology transfer ● **Industrial and Economic Impact** ■ Technology Leadership ✓ Establishes Korea as global leader in ultra small formfactor optical transceiver for AI data center ✓ First prototype of optical transceivers using optical waveguide embedded glass interposer